

3. (a) When a mixture of zinc and copper(II) oxide is heated it ignites and burns brightly.
When silver is heated with copper(II) oxide no reaction takes place.

- (i) List silver, copper and zinc in order of reactivity. [1]

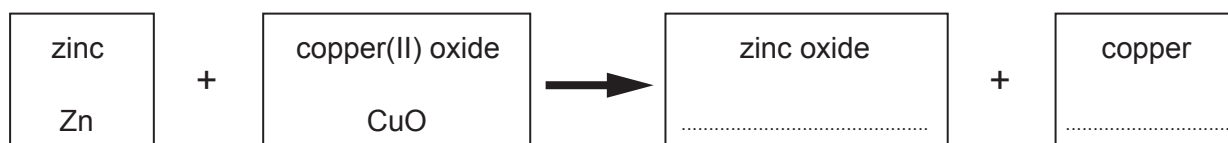
Most reactive

.....

Least reactive

- (ii) The boxes below show the word equation and an incomplete symbol equation for the reaction taking place between zinc and copper(II) oxide.

Complete the symbol equation. [2]



- (b) A **solid** precipitate of silver chloride is formed when silver nitrate solution is added to sodium chloride solution.

- (i) Complete the equation for this reaction by writing the **state symbol** for the silver chloride formed in the brackets. [1]



- (ii) Underline the correct word(s) in the bracket to complete the sentence. [1]

The reaction shows that silver chloride is (**soluble** / **insoluble** / **a mixture**) in water.

